

Planning Report

An Open Project in Deep Learning: Hand Gesture Recognition System

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Project Goal

The project goal is to develop a hand gesture recognition system.

Secondary goals if primary goal is achieved:

- Enable custom gesture addition without largely retraining the network.
- Investigates if contrastive learning works better than transfer learning on embedding model pre-trained via supervised learning methods in terms of accuracy and efficiency.
- Investigates if our approach could possibly utilize a large labelled dataset for contrastive learning to minimize fine-tuning efforts given a relatively small gesture dataset.
- Match and annotate gestures from videos.

Significance

While most common gesture recognition models focuses on a fixed set of gestures, this project aims to improve the flexibility and usability of gesture recognition systems, with the following sub-goals:

1. Address the need of custom gestures in some applications.
2. Automate the annotation process for unannotated videos and allow efficient auto-annotation of certain unseen gestures into the database.

Real Scenario of Application:

1. Quick adaptation of the system to the controls of games, screen control (zooming, etc), sign language (gloss extraction based on similar gestures), etc with minimal re-training / finetuning if possible.
2. Sign language annotation.

Problem statement

The difficulty of the task of hand gesture recognition can basically be broken down into several parts:

1. Static vs Dynamic:
 - While nowadays AI models are advancing quickly, the hand gesture recognition tasks have also evolved. Instead of recognizing static postures, there are high demands on high quality recognition on dynamic gestures.

2. Scarcity of High-Quality Well-Annotated Data:
 - For some applications, it might be costly or not cost-effective to collect high-quality well-labelled gesture data. An example being sign language recognition on a relatively minor language. In such cases, it might be easier to provide unannotated samples for training or fine-tuning. One of the final goals of the project, if everything runs smoothly, is that the method should work, even when provided relatively few samples for reference.

Previous works:

1. TwoStreamNetwork (SLRT) and SlowFastSign for sign language
2. On-Device Real-Time Custom Hand Gesture Recognition (<https://arxiv.org/pdf/2309.10858>)
3. Temporal Contrastive Learning (<https://arxiv.org/pdf/2101.07974>)

Proposed solutions (deliverables)

- An easy-to-transfer gesture recognition model that can utilize a relatively small, previously unseen set of hand gesture videos, and with minimal retraining or fine-tuning, annotate the set of gestures from videos, including livestreams.
- Utilize MediaPipe to extract the keypoints of the hands to allow a higher variety of augmentations for training.

Proposed timelines

- September
 - Study foundation papers related to the target problem and decide the basic directions of the project.
- October
 - Study papers related to the architectures and methodologies that could be used to improve the efficiency.
 - Train models on smaller datasets to test if the concept is feasible.
 - Focuses mainly on static gestures.
 - Modify the directions of the project.
- November
 - Perform more detailed analysis on the methods and their accuracies and efficiencies.
 - Explore the weaknesses of the methods.
- December – January
 - Explore on dynamic gestures.
- February – March
 - Analysis on the accuracies and efficiencies of the methods.
 - Explore the possibilities of auto-annotation of sign language videos for previously unseen hand gestures.